MICROBIOLOGY · MCQ SET

# E. coli, Klebsiella, Proteus & Enterobacter spp.

SGL 1 · Dr Samira Fattah · GUE BLOCK · PRACTICE MCQs



20 questions, mixed difficulty, 9 case-based. Attempt every question before checking the answer key on the last page.

| 4 | 10 | 6 |
|---|---|---|
| Easy | Medium | Hard |

1. EASY

**Which order do E. coli, Klebsiella, Proteus and Enterobacter all belong to?**

A. Pseudomonadales
B. Enterobacterales
C. Lactobacillales
D. Bacillales
E. Vibrionales

2. EASY

**Which single biochemical test result is shared by all four organisms?**

A. Oxidase negative
B. Urease positive
C. Lactose negative
D. Indole positive
E. Citrate negative

3. EASY

**What is the classic colony appearance of Klebsiella pneumoniae on culture?**

A. Swarming with concentric rings
B. Dry, flat, non-lactose-fermenting
C. Mucoid colonies, often string-test positive
D. Green-pigmented colonies
E. Tiny pinpoint colonies only on blood agar

4. EASY

**What is nitrofurantoin's approved scope of use for these urinary pathogens?**

A. Pyelonephritis only
B. Bladder infection (cystitis) only
C. Urosepsis only
D. Prostatitis only
E. Any site, any severity

5. MEDIUM CASE

A 45-year-old woman has fever, flank pain and costovertebral angle tenderness. Imaging shows a dilated, obstructed collecting system on the affected side.

**Which single finding on examination would prompt urgent urological assessment?**

A. Dysuria and suprapubic discomfort alone
B. Urine culture showing >10^5 CFU/mL of E. coli
C. Fever with flank pain AND obstruction on imaging
D. Mild urgency without fever
E. A positive nitrite dipstick

6. MEDIUM

**Which organism's UPEC pathotype designation CANNOT be established by colony appearance on MacConkey agar alone?**

A. Klebsiella pneumoniae
B. Proteus mirabilis
C. Escherichia coli
D. Enterobacter cloacae complex
E. All four equally

7. MEDIUM CASE

An 80-year-old man with a long-term indwelling catheter has a urine culture growing Klebsiella pneumoniae at 10^5 CFU/mL. He is afebrile with no new urinary symptoms, abdominal pain or confusion.

**What is the single best next step?**

A. Start antibiotics immediately
B. No antibiotics — this is asymptomatic bacteriuria
C. Repeat culture tomorrow
D. Start fosfomycin as prophylaxis
E. Refer for urology work-up

**8. MEDIUM**

**A urease-strong-positive, swarming, non-lactose-fermenting Gram-negative rod is most likely to be:**

A. Escherichia coli

B. Klebsiella pneumoniae

C. Proteus mirabilis

D. Enterobacter cloacae complex

E. Cannot determine from these features

**9. MEDIUM CASE**

A patient develops ascending infection that begins with periurethral colonization and progresses to bladder colonization despite normal voiding.

**Which virulence mechanism best explains how the bacteria resist being washed out by urine flow before ascending further?**

A. Hemolysin-mediated cell lysis

B. LPS-triggered inflammation

C. Fimbrial adhesion to uroepithelial cells

D. AmpC β-lactamase production

E. Urease-driven pH change

**10. MEDIUM**

**A plain KUB X-ray shows a large branching calculus filling the renal pelvis and calyces. Which organism is most strongly associated with this finding?**

A. Escherichia coli

B. Klebsiella pneumoniae

C. Proteus mirabilis

D. Enterobacter cloacae complex

E. Any Enterobacterales equally

**11. MEDIUM CASE**

A patient with a complicated UTI has had two prior urine cultures with different resistance patterns and recent antibiotic exposure.

**What should guide the choice of empirical antibiotic in this situation?**

A. The site of infection alone

B. Severity alone

C. Site, severity, prior cultures/antibiotics and the local antibiogram together

D. Patient preference only

E. Whichever antibiotic was used last time, regardless of past susceptibility

**12. MEDIUM**

**A Klebsiella pneumoniae urinary isolate is reported susceptible to ampicillin. What is the most appropriate interpretation?**

A. Expected, since Klebsiella is usually ampicillin-susceptible

B. Unexpected — Klebsiella has expected resistance to ampicillin; verify the result

C. Irrelevant, since ampicillin is never used for UTI

D. Confirms the isolate is an ESBL producer

E. Confirms the isolate is Proteus, not Klebsiella

**13. MEDIUM CASE**

A patient with invasive Enterobacter cloacae complex bacteremia initially improves on ceftriaxone, then relapses on the same drug several days later.

**What does this clinical course suggest, and what should guide the next antibiotic choice?**

A. Treatment failure from an ESBL enzyme present from the start

B. AmpC overproduction induced during cephalosporin therapy — avoid ceftriaxone, consider cefepime or a carbapenem

C. A new, unrelated infection unrelated to antibiotic exposure

D. Drug allergy — switch to a different drug class entirely regardless of mechanism

E. Laboratory error; repeat the same antibiotic

**14. MEDIUM**

**Which patient group is a recognized EXCEPTION to the rule that asymptomatic bacteriuria should not be treated?**

A. Older adults in long-term care

B. Patients with diabetes

C. Patients with chronic indwelling catheters

D. Pregnant patients

E. Patients with a single prior UTI

**15. HARD CASE**

A patient with a long-term urinary catheter has asymptomatic bacteriuria on routine culture, but now develops new catheter blockage and visible encrustation.

**What is the most appropriate interpretation and action?**

A. Her symptoms are definitely unrelated to the isolate; no action needed

B. The isolate is clinically insignificant since she is afebrile

C. Fosfomycin should be started empirically regardless of symptoms

D. Catheter-related problems and infection both require clinical assessment — bacteriuria alone does not mandate antibiotics, but new symptoms or device malfunction do

E. She should have the catheter removed immediately without further assessment

**16. HARD CASE**

A patient with recurrent UTIs is found to have a staghorn calculus on imaging.

**Which organism's virulence mechanism directly explains this radiographic finding, distinct from the other three organisms discussed?**

A. E. coli fimbrial adhesion

B. Klebsiella capsular polysaccharide

C. Proteus urease-driven alkalinization and struvite precipitation

D. Enterobacter AmpC β-lactamase

E. None of these mechanisms relate to stone formation

**17. HARD**

**Which statement about the four organisms' indole test results is CORRECT?**

A. All four are indole positive

B. E. coli is indole positive; Klebsiella and Proteus are indole negative; Enterobacter is usually indole negative

C. Proteus is indole positive; the rest are negative

D. Klebsiella is indole positive; the rest are negative

E. Indole testing cannot distinguish any of the four organisms

**18. HARD CASE**

A urine isolate is lactose-fermenting, nonmotile, citrate-positive, urease-positive and indole-negative, with mucoid colonies.

**Which organism best fits this isolate, and what is the key antibiotic implication?**

A. E. coli — treat confidently with nitrofurantoin for any site

B. Klebsiella pneumoniae — assume ampicillin resistance and consider ESBL risk

C. Proteus mirabilis — nitrofurantoin remains first-line

D. Enterobacter — ceftriaxone is the preferred agent

E. Cannot be an Enterobacterales member

**19. HARD CASE**

A pregnant woman at a routine antenatal visit has a screening urine culture positive for E. coli at a significant colony count, with no urinary symptoms.

**What is the most appropriate next step in her management?**

A. No treatment, since she has no fever or flank pain

B. Treat the asymptomatic bacteriuria, since pregnancy is a recognized exception

C. Treat only if she develops symptoms

D. Treat with nitrofurantoin regardless of trimester without further thought

E. Delay all testing until after delivery

**20. HARD**

**A urinary isolate is lactose negative, highly motile with swarming, urease strongly positive and indole negative. Which single antibiotic class/agent should specifically be AVOIDED for this organism, based on the lecture?**

A. Fosfomycin

B. TMP-SMX

C. Nitrofurantoin

D. A carbapenem

E. Cefepime

# ANSWER KEY & RATIONALE

*Don't peek until you've committed to an answer on every question.*

| Q | Ans | Why |
|---|---|---|
| 1 | B | All four organisms belong to the order Enterobacterales. |
| 2 | A | All four are usually oxidase negative and reduce nitrate — the shared Enterobacterales feature stated in the lecture. |
| 3 | C | Klebsiella colonies are often mucoid and some isolates are string-test positive (viscous string lifted with a loop). |
| 4 | B | Nitrofurantoin is for bladder infection (cystitis) only — it must not be used for pyelonephritis or bacteremia. |
| 5 | C | Fever + flank pain with obstruction on imaging is urosepsis-risk territory and requires urgent urological assessment to restore drainage. |
| 6 | C | Colony appearance on MacConkey cannot establish the UPEC pathotype of an E. coli isolate — this is stated explicitly for E. coli in the lecture. |
| 7 | B | Long-term catheter bacteriuria alone is not an indication for antibiotics in an asymptomatic patient. |
| 8 | C | Strong urease positivity plus swarming plus non-lactose-fermenting is the classic Proteus mirabilis profile. |
| 9 | C | Fimbriae/adhesion help bacteria resist removal by urine flow, allowing colonization and further ascent — stated directly under 'Ascending urinary tract infection'. |
| 10 | C | Staghorn calculi form from struvite precipitation driven by Proteus urease activity — the organism most strongly linked to this finding. |
| 11 | C | Empirical selection should weigh site, severity, prior cultures/antibiotics and the local antibiogram together, per the lecture's empirical selection framework. |
| 12 | B | Klebsiella pneumoniae has expected resistance to ampicillin — an ampicillin-susceptible result is unexpected and should be verified. |
| 13 | B | This pattern — initial response then relapse on the same cephalosporin — is classic for AmpC overproduction induced during therapy; ceftriaxone should generally be avoided for invasive Enterobacter infection in favor of cefepime or a carbapenem. |
| 14 | D | Pregnancy is the key exception (along with urological procedures with mucosal trauma) where asymptomatic bacteriuria is actively treated. |
| 15 | D | Catheter problems (blockage, encrustation) and infection both warrant clinical assessment; bacteriuria alone does not mandate antibiotics, but a malfunctioning, encrusted catheter is a different problem requiring action. |
| 16 | C | Proteus urease hydrolyzes urea to ammonia, alkalinizing urine and precipitating struvite/carbonate apatite — the mechanism specific to staghorn calculus formation among these four organisms. |
| 17 | B | E. coli is indole positive; Klebsiella and Proteus are indole negative; Enterobacter is usually indole negative — per the comparative identification table. |
| 18 | B | Lactose+, nonmotile, citrate+, urease+, indole− with mucoid colonies fits Klebsiella pneumoniae; expected ampicillin resistance and ESBL risk should be assumed. |
| 19 | B | Pregnancy is a recognized exception where asymptomatic bacteriuria is screened for and treated, regardless of absent symptoms. |
| 20 | C | This profile (non-lactose-fermenting, swarming, strongly urease-positive, indole-negative) is Proteus, for which nitrofurantoin is specifically described as unsuitable. |